

Solution Code - Module 3-Core Java - InheritanceExample

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q18_InheritanceExample.java

```
1 | // Base class Animal
2 | class Animal {
3 |     public void makeSound() {
4 |         System.out.println("Animal makes a generic sound");
5 |     }
6 | }
7 |
8 | // Subclass Dog extending Animal
9 | class Dog extends Animal {
10 |     // Overriding the makeSound method
11 |     @Override
12 |     public void makeSound() {
13 |         System.out.println("Bark");
14 |     }
15 | }
16 |
17 | public class Q18_InheritanceExample {
18 |     public static void main(String[] args) {
19 |         // Instantiate both classes
20 |         Animal genericAnimal = new Animal();
21 |         Dog myDog = new Dog();
22 |
23 |         // Call their methods
24 |         System.out.print("Calling Animal.makeSound(): ");
25 |         genericAnimal.makeSound();
26 |
27 |         System.out.print("Calling Dog.makeSound(): ");
28 |         myDog.makeSound();
29 |
30 |         // Polymorphism demo
31 |         System.out.print("Calling Polymorphic Animal.makeSound() (assigned to a Dog instance): ");
32 |         Animal polyDog = new Dog();
33 |         polyDog.makeSound();
34 |     }
35 | }
```